

Multiple Linear Regression (MLR)

Outline

- What is MLR?
- Adjusted R^2
- Example

Why MLR?

- Real-world engineering example
- Multiple factors affect output
- Need mathematical model
- Example:
 - **Concrete strength:** Strength depends on **cement, water, aggregates, and curing time**, multiple inputs determine output strength.
 - **Fuel efficiency:** Mileage depends on **engine size, vehicle weight, aerodynamics, and driving conditions**.
 - **Load prediction:** Structural load depends on **material properties, span length, cross-section, and external forces**.

Data for Multiple Linear Regression Models

	SLR		MLR				
	X	Y	X_1	X_2	$\dots$	X_p	Y
case 1 :	x_1	y_1	x_{11}	x_{12}	$\dots$	x_{1p}	y_1
case 2 :	x_2	y_2	x_{21}	x_{22}	$\dots$	x_{2p}	y_2
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
case n :	x_n	y_n	x_{n1}	x_{n2}	$\dots$	x_{np}	y_n

- ▶ For SLR, we observe **pairs** of data values.
- ▶ For MLR, we observe **rows** of data values.
- ▶ Each row (or pair) is called a **case**, a **record**, or a **data point**
- ▶ y_i is the **response** (or **dependent variable**) of the i th case
- ▶ There are p **explanatory variables** (or **predictors**, **covariates**), and x_{ik} is the value of the explanatory variable X_k of the i th case

This is not a regression line any longer, but a *regression surface* and we relate y to more than one predictor variable $x_1, x_2, \dots, x_p$.

(ex. Blood sugar level vs. weight and age)

Multiple Linear Regression Models

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i$$

In the model above,

- ▶ ε_i 's (errors, or noise) are i.i.d. $N(0, \sigma^2)$
- ▶ Parameters include:
 - ▶ β_0 = intercept;
 - ▶ β_k = regression coefficient (slope) for the k th explanatory variable, $k = 1, \dots, p$
 - ▶ $\sigma^2 = \text{Var}(\varepsilon_i)$ = the variance of errors
- ▶ Observed (known): $y_i, x_{i1}, x_{i2}, \dots, x_{ip}$
Unknown: $\beta_0, \beta_1, \dots, \beta_p, \sigma^2, \varepsilon_i$'s
- ▶ Random: ε_i 's, y_i 's
Constants (not random): β_k 's, σ^2, x_{ik} 's

Multiple Linear Regression Models in Matrix Notation

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{Y_{n \times 1}} = \underbrace{\begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{p1} \\ 1 & x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & x_{2n} & \cdots & x_{pn} \end{bmatrix}}_{X_{n \times (p+1)}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}}_{\beta_{(p+1) \times 1}} + \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}}_{\varepsilon_{n \times 1}}$$

or

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon$$

MODEL FIT

$$R^2 = 1 - \frac{SSE}{SST}$$

Explain:

- Total variation
- Explained variation
- Residual variation
- Interpretation (0 to 1)

$$R^2$$

Unfortunately, there is a problem with R^2 : **Its value can be inflated** by adding lots of predictors into the model even if most of these predictors are frivolous.

For example, suppose y is the sale price of a house. Then sensible predictors include

x_1 = the interior size of the house,

x_2 = the size of the lot on which the house sits,

x_3 = the number of bedrooms,

x_4 = the number of bathrooms, and

x_5 = the house's age.

Now suppose we add in

x_6 = the diameter of the doorknob on the coat closet,

x_7 = the thickness of the cutting board in the kitchen,

x_8 = the thickness of the patio slab.

Why Adjusted R^2 ?

Problem:

- R^2 always increases when adding variables

Adjusted R^2

$$R_a^2 = 1 - \frac{SSE / (n - (p + 1))}{SST / (n - 1)} = 1 - \frac{n - 1}{n - (p + 1)} \times \frac{SSE}{SST}$$

Where:

SSE = Residual Sum of Squares

n = number of observations

p = number of predictors

Explain:

- Penalizes unnecessary predictors
- Better for multiple regression

Difference Between R^2 and Adjusted R^2

R^2	Adjusted R^2
Always increases	Can decrease
Does not penalize	Penalizes extra variables
Simple measure	More reliable

Example

- Calculate the coefficients β_0 , β_1 and β_2 using **Multiple Linear Regression**.

x_1	x_2	y
1	2	3
2	1	5
3	4	7

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

$$\beta = (X^T X)^{-1} X^T Y$$

Intercept

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 4 \end{bmatrix}, \quad Y = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 4 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 3 & 6 & 7 \\ 6 & 14 & 16 \\ 7 & 16 & 21 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 15 \\ 34 \\ 39 \end{bmatrix}$$

we need to calculate the **inverse** of the matrix $X^T X$:

$$X^T X = \begin{bmatrix} 3 & 6 & 7 \\ 6 & 14 & 16 \\ 7 & 16 & 21 \end{bmatrix}$$

The inverse of a 3x3 matrix can be calculated using the formula:

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{Adj}(A)$$

Where:

- $\det(A)$ is the determinant of the matrix,
- $\text{Adj}(A)$ is the adjugate (or adjoint) matrix, which is the transpose of the cofactor matrix.

$$X^T X = \begin{bmatrix} 3 & 6 & 7 \\ 6 & 14 & 16 \\ 7 & 16 & 21 \end{bmatrix}$$

$$\det(X^T X) = 3 \begin{vmatrix} 14 & 16 \\ 16 & 21 \end{vmatrix} - 6 \begin{vmatrix} 6 & 16 \\ 7 & 21 \end{vmatrix} + 7 \begin{vmatrix} 6 & 14 \\ 7 & 16 \end{vmatrix}$$

$$\begin{aligned} \det(X^T X) &= 3(14 \cdot 21 - 16 \cdot 16) - 6(6 \cdot 21 - 16 \cdot 7) + 7(6 \cdot 16 - 14 \cdot 7) \\ &= 3(38) - 6(14) + 7(-2) = 114 - 84 - 14 \\ &= \boxed{16} \end{aligned}$$

Let $A = X^T X = \begin{bmatrix} 3 & 6 & 7 \\ 6 & 14 & 16 \\ 7 & 16 & 21 \end{bmatrix}$

Cofactor Matrix

$$C_{ij} = (-1)^{i+j} M_{ij}$$

where M_{ij} is the minor.

Compute Minors & Cofactors

$$C_{11} = \begin{vmatrix} 14 & 16 \\ 16 & 21 \end{vmatrix} = 14(21) - 16(16) = 38$$

$$C_{12} = - \begin{vmatrix} 6 & 16 \\ 7 & 21 \end{vmatrix} = -(126 - 112) = -14$$

$$C_{13} = \begin{vmatrix} 6 & 14 \\ 7 & 16 \end{vmatrix} = 96 - 98 = -2$$

$$C_{21} = - \begin{vmatrix} 6 & 7 \\ 16 & 21 \end{vmatrix} = -(126 - 112) = -14$$

$$C_{22} = \begin{vmatrix} 3 & 7 \\ 7 & 21 \end{vmatrix} = 63 - 49 = 14$$

$$C_{23} = - \begin{vmatrix} 3 & 6 \\ 7 & 16 \end{vmatrix} = -(48 - 42) = -6$$

$$C_{31} = \begin{vmatrix} 6 & 7 \\ 14 & 16 \end{vmatrix} = 96 - 98 = -2$$

$$C_{32} = - \begin{vmatrix} 3 & 7 \\ 6 & 16 \end{vmatrix} = -(48 - 42) = -6$$

$$C_{33} = \begin{vmatrix} 3 & 6 \\ 6 & 14 \end{vmatrix} = 42 - 36 = 6$$

Final Cofactor Matrix is

$$\begin{bmatrix} 38 & -14 & -2 \\ -14 & 14 & -6 \\ -2 & -6 & 6 \end{bmatrix}$$

Adjugate (Adjoint) Matrix

- Adjugate is the **transpose of cofactor matrix**:

$$\text{Adj}(A) = \begin{bmatrix} 38 & -14 & -2 \\ -14 & 14 & -6 \\ -2 & -6 & 6 \end{bmatrix}$$

Compute Inverse

$$(X^T X)^{-1} = \frac{1}{16} \begin{bmatrix} 38 & -14 & -2 \\ -14 & 14 & -6 \\ -2 & -6 & 6 \end{bmatrix}$$

Divide each element by 16:

$$(X^T X)^{-1} = \begin{bmatrix} 2.375 & -0.875 & -0.125 \\ -0.875 & 0.875 & -0.375 \\ -0.125 & -0.375 & 0.375 \end{bmatrix}$$

Multiply by $X^T Y$

$$\overset{2}{X^T Y} = \begin{bmatrix} 15 \\ 34 \\ 39 \end{bmatrix}$$

Now compute:

$$\beta = \begin{bmatrix} 2.375 & -0.875 & -0.125 \\ -0.875 & 0.875 & -0.375 \\ -0.125 & -0.375 & 0.375 \end{bmatrix} \begin{bmatrix} 15 \\ 34 \\ 39 \end{bmatrix}$$

$\overset{1}$ $\overset{2}$

$$2.375(15) - 0.875(34) - 0.125(39) = 35.625 - 29.75 - 4.875 = 1$$

$$-0.875(15) + 0.875(34) - 0.375(39) = -13.125 + 29.75 - 14.625 = 2$$

$$-0.125(15) - 0.375(34) + 0.375(39) = -1.875 - 12.75 + 14.625 = 0$$

$$\beta = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$y = 1 + 2x_1 + 0x_2$$

Or

$$y = 1 + 2x_1$$

This means, x_2 does not contribute after accounting for x_1

Summary

- **In Simple Regression:**
 - R^2 measures variation explained by one predictor.
- **In Multiple Regression:**
 - R^2 measures variation explained by **all predictors together**.

In Multiple Regression, we usually prefer:

- Adjusted R^2 Because:
 - Adding **more variables** always increases R^2
 - Even if those variables are **useless**
- Adjusted R^2 **penalizes unnecessary predictors.**

In multiple regression:

- R^2 never decreases when you add variables.
- Adjusted R^2 can decrease.
- That's why Adjusted R^2 is more reliable.

Summary

- Step 1: Model Setup
 - Form the Design Matrix (X)
 - Form the Response Vector (Y)
- Step 2: Compute the Transpose of X
 - Compute $X^T X$
- Step 3: Compute Matrix Inverse
 - Compute the **Determinant** of $X^T X$
 - Find the **Cofactor Matrix**
 - Find the **Adjugate (Adjoint) Matrix**
 - Compute the Inverse $(X^T X)^{-1}$
- Step 4: Compute Coefficients
 - Compute $X^T Y$
 - Multiply $(X^T X)^{-1}$ and $X^T Y$
- Step 5: Form the Regression Equation
 - Substitute β values into:

$$\beta = (X^T X)^{-1} X^T Y$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$$

Assignment

x_1	x_2	y
1	1	6
1	2	8
2	1	9
2	2	11